



2.3.4 Wrap-Up: Reflection

1. How many of the UNO proofs did the class correctly complete?
2. Which steps in the UNO proof were the easiest to complete?
3. Which steps in the UNO proof were hardest to complete?



Unit 2, Lesson 4: Hypothesis and Conclusion



Benchmark

MA.912.LT.4.3 Identify and accurately interpret “if...then,” “if and only if,” “all” and “not” statements. Find the converse, inverse, and contrapositive of a statement.



Learning Targets

- ☐ I can identify the hypothesis of a statement.
- ☐ I can identify the conclusion of a statement.
- ☐ I can write the converse, inverse, and contrapositive of a conditional statement.



2.4.1 Warm-Up: Prototype Fabricator

Jeff Tiedeken is a prototype fabricator. He creates metal prototypes through bending, cutting, welding, and assembling.



Jeff states that most people who come in for help do not have an idea of what they need, so his first job is to understand the problem.

1. How does understanding the problem help?

Jeff said that once he understands the problem, he then solves the problem with other people.

2. Why do you think discussing a problem with others can be helpful?



2.4.2 Guided Instruction: Hypothesis and Conclusion

Often information is given as an if-then statement. A few examples are shown.

- If you leave the lights on, then it will take longer for people to fall asleep.
- If a goldfish is not exposed to light, then it will lose its color.
- If a quadrilateral has 4 right angles, then it is a rectangle.

1. Write a new if-then statement using a different phrase following "if."

If _____, then it
will take longer for people to fall asleep.

2. Write a new if-then statement using a different phrase following "then."

If a quadrilateral has 4 right angles, then _____
_____.

A **conditional statement** is a statement in the format of "If P , then Q ." The phrase P is the hypothesis. The phrase Q is the conclusion.

3. What is the hypothesis for “If a quadrilateral has 4 right angles, then it is a rectangle”?

4. Write a conclusion for each hypothesis.

a. If yesterday was Sunday, then _____.

b. If you see lightning, then _____.

5. William Camden wrote, “The early bird catches the worm.” Rewrite Camden’s statement as a conditional statement.



2.4.3 Exploration: Switching the Hypothesis and Conclusion

1. Draw a square or a triangle.

2. If you drew a square, then draw a circle inside the square.

If you drew a triangle, then draw a circle outside the triangle.

3. Discuss with your partner what happens when the hypothesis and conclusion of each conditional are switched.

- If you drew a circle inside the square, then draw a square.
- If you drew a circle outside the triangle, then draw a triangle.

Summarize your discussion.



2.4.4 Guided Instruction: Converse, Inverse, and Contrapositive

The **converse statement** is a statement in which the given conditional, "If P , then Q ," changes to "If Q , then P ."

1. Write the converse of "If it is raining, then the grass is wet."
2. Write a counterexample that shows the converse is not true.

The **inverse statement** is a statement in which the given conditional, "If P , then Q ," changes to "If not P , then not Q ."

3. Complete the inverse for "If it is raining, then the grass is wet."
If it is _____ raining, then the grass is _____ wet.
4. Write a counterexample that shows the inverse is not true.

The **contrapositive statement** is a statement in which the given conditional, "If P , then Q ," changes to "If not Q , then not P ."

5. Write the contrapositive for "If it is raining, then the grass is wet."



2.4.5 Guided Practice: Converse, Inverse, and Contrapositive

Write the converse, inverse, and contrapositive for each conditional.

1. "If you're not making mistakes, then you're not doing anything." (Quote by John Wooden, an American basketball player and coach)

Converse	
Inverse	
Contrapositive	

2. "If you think you're always in control, then you're not going fast enough." (Quote by Mario Andretti)

Converse	
Inverse	
Contrapositive	



2.4.6 Wrap-Up: Cool Down

1. Rewrite the following quote by Michelle Obama as a conditional statement.

"When someone is cruel or acts like a bully, you don't stoop to their level."

2. Complete the table for the conditional statement shown.

If you have ridden the roller coaster SheiKra, then you have been to Busch Gardens.

Hypothesis	
Conclusion	

3. Write the converse, inverse, and contrapositive for the conditional statement shown.

If a fruit is yellow, then it is a banana.

Converse	
Inverse	
Contrapositive	